EPSR-test-ex123 Lattice refinement of a triclinic crystal (with Bragg peaks)

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the lattice-refinement loop of the `epsr` engine [1,2,3], a random search over the unit-cell parameters a, b, c, alpha, beta, gamma that keeps a trial lattice when the fit to the diffraction data improves. It is the route by which EPSR refines the cell of a crystalline sample together with its structure.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (do `while (ilatrefine.lt.nlatrefine+1)` loop, R-factor, accept/reject), `gensq`
- `ato_manipulation_rev4.f`: `change_boxsize`, `atogen`, `bragginten`
- `auxiliary_routines.f`: `convertphibthetacphic`, `convertalphabetagamma`; `ato_read_and_save.f`: `(un)normalise_coordinates`

Rust (port, rust/epsr-core/src/):

- `bragg.rs`: `structure_and_lattice`, `change_boxsize`, `convertphibthetacphic`, `bragginten`
- `tools/makelattice.rs`: `convertalphabetagamma`; `ato.rs`: `normalise_coordinates`, `unnormalise_coordinates`; `rng.rs`: `ran1`

Physical system

The triclinic S/O1 crystal of ex121 (588 tethered atoms, 7 × 7 × 6 cells of a, b, c = 4.3, 4.8, 5.4 Å, alpha, beta, gamma = 80°, 100°, 110°, 300 K); EPSR-test `.pcof`, weights and data used unchanged as numerical input. With Bragg peaks [4] the R-factor against the three data sets depends on the lattice; trial cells are drawn with step sizes 0.2 Å and 1° and rescaled to constant density.

Inputs

`scripts/make_case.sh` regenerates the inputs (verified byte-identical). Keywords changed: `iinit 1`, `ntimes 2`, `qwidthqmax 0.02 3.0`, `mplicities 7 7 6`, `hklqmin 0.0 8.0 0.05`, `nlatrefine 3`, `latrefine 0.2 0.2 0.2 1.0 1.0 1.0`. `commands.txt`: `repeat 2`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Per the README, in pass 1 the trials of steps 1 and 2 are accepted and step 3 rejected; in pass 2 steps 1 and 3 are rejected and step 2 accepted. The refined lattice is written to the `.ato` (stored Fortran reference: 30.224 × 33.667 × 32.641 Å box edges) and read back in pass 2. Each step runs `atogen` + `gensq` + Bragg correction and six `ran1` calls, so any difference would shift the random sequence.

Notes

Reproduced: `clatnew` is rescaled with `blatfac`; when the last trial is rejected the box is restored but S(Q) is not recomputed.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] G. L. Squires, Introduction to the Theory of Thermal Neutron Scattering, Cambridge University Press (1978).
[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.